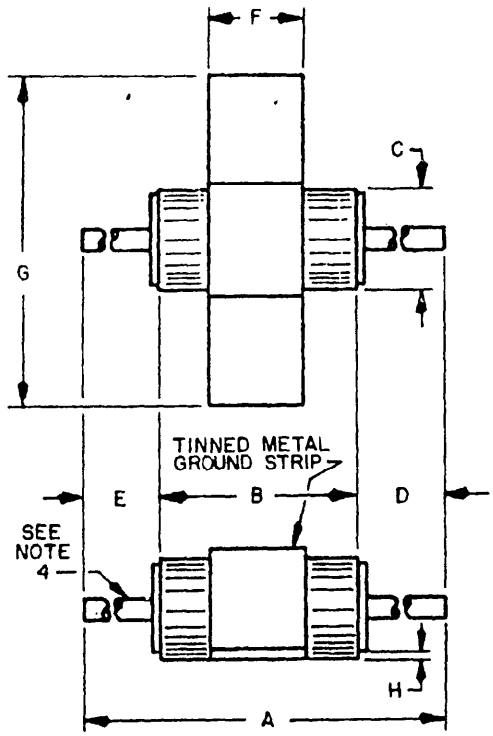
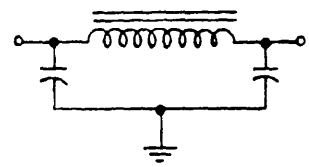
MILITARY SPECIFICATION SHEET

FILTER, RADIO INTERFERENCE,
STYLE FL49

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter described herein shall consist of this document and the latest issue of Specification MIL-F-15733.





TYPICAL CIRCUIT DIAGRAM

TABLE I. Filter dimensions.

Type designation 1/ 2/	Dash number 3/	A 4/		B		C		D		E		F		G 4/		H	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
FL49GW1ED3	0001	1.049 (26.64)	1.111 (28.22)	.240 (.610)	.260 (.660)	---	Dia .125 (3.18)	.405 (10.29)	.425 (10.80)	.405 (10.29)	.425 (10.80)	.105 (2.67)	.125 (3.15)	.390 (9.91)	.410 (10.41)	---	.210 (5.33)

1/ The letter "W" indicates the insertion loss versus frequency characteristics specified in table II.
2/ The letter "D" indicates the operating temperature range -55°C to +125°C.
3/ Preferred for use in the design of equipment.
4/ Metric equivalents are in parentheses.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Circuit diagram is for information only.
4. Leads shall be solid, tinned, AWG 20 .032 (.81 mm) diameter.

FIGURE 1. Filter configuration.

FSC 5915

REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Weight: 0.7 gram.

Case: Metallic mounting strap.

Terminals: Solderable.

Operating temperature range:
-55° to +125°C; temperature characteristic D.

Rated voltage: 70 Vdc.

Rated current: 10 amperes, dc.

DC resistance: 0.01 ohm maximum.

Insertion-loss characteristic: Symbol W (see table II).

Seal: Not applicable.

Capacitance to ground: Method 305, MIL-STD-202. Test frequency: 1,000 Hz ±100 Hz. Measured capacitance shall be at least 12,000 pF.

Temperature rise: 15°C maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202. The test voltage of 200 Vdc shall be applied for 1 to 5 seconds. Charging current shall not exceed 50 mA.

Barometric pressure (reduced): Method 105, test condition B (50,000 ft), MIL-STD-202.
Dielectric withstanding voltage: In accordance with initial requirements, except that test voltage shall be 1.2 times rated dc voltage.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Insulation resistance measured at 25°C between both terminals connected together and the case shall be at least 1,000 megohms.

Voltage drop: Not applicable.

Insertion-loss: In accordance with MIL-F-15733 and table II.

Overload: In accordance with MIL-F-15733.
Measurements at +25°C after test:
Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition A, MIL-STD-202.
Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

TABLE II. Insertion-loss versus frequency.

Dash number	Minimum no. load insertion-loss (dB) in accordance with MIL-STD-220, at +25°C					
	10 MHz	20 MHz	100 MHz	400 MHz	1 GHz	10 GHz
0001	25	40	70	65	65	65

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion:
Thermal shock: Method 107, test condition A, MIL-STD-202, except the temperature at step 3 shall be +125°C.
Immersion: Method 104, test condition A, MIL-STD-202.
Measurements after final cycle:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.
Insulation resistance shall be not less than 1,000 megohms.
Insertion-loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.
Electrical load: 100 percent of the rated voltage between terminals and case.

Vibration, high frequency: Method 204, test condition D (20 G), MIL-STD-202.
Electrical load: Rated voltage and rated current shall be applied continuously during test.
Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

Moisture resistance: Not applicable.

Life: Method 108, test condition D, MIL-STD-202. Test temperature +125°C.
Test voltage: 1.4 times the rated voltage.
Test current: 10 amperes.
Measurements after test:
Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.
Insulation resistance shall be not less than 1,000 meghoms.
Insertion-loss (check test) shall be measured and shall meet the initial requirements.

Part number: M15733/64- (dash number from table I).

Marking: Filters shall not be marked. Full marking, in accordance with MIL-F-15733, shall be marked on the unit package.

Supersession data: Filters in this specification supersede filters in MIL-F-15733/50 as follows:

MIL-F-15733/50 Part number	MIL-F-15733/64 Part number
M15733/50-0003	M15733/64-0001

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - AR, MI, GL
Navy - SH
Air Force - 85, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV, WC
Navy - CG, MC, AS, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5915-0197)

6. Mounting hardware (lockwasher and nut) shall be supplied with filter.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-F-15733/64 26 JUNE 74

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802